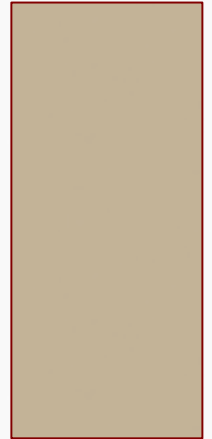


GAME DESIGN PRINCIPLES PART I

ADVANCED GAME DESIGN & DEVELOPMENT

DANNY JUGAN



GAME DESIGN PRINCIPLES

- Most of these principles are common sense
- So why are we going over them?
 - 1 – People (game designers) don't always do what is common sense. You would think they do because it's common sense, but you'd be surprised how often they don't.
 - 2 – It's important to have specific reasons (and research) behind the decisions you're making for your game.

AFFORDANCE CUES

- The physical qualities of UI elements communicate how a player should interact with them.

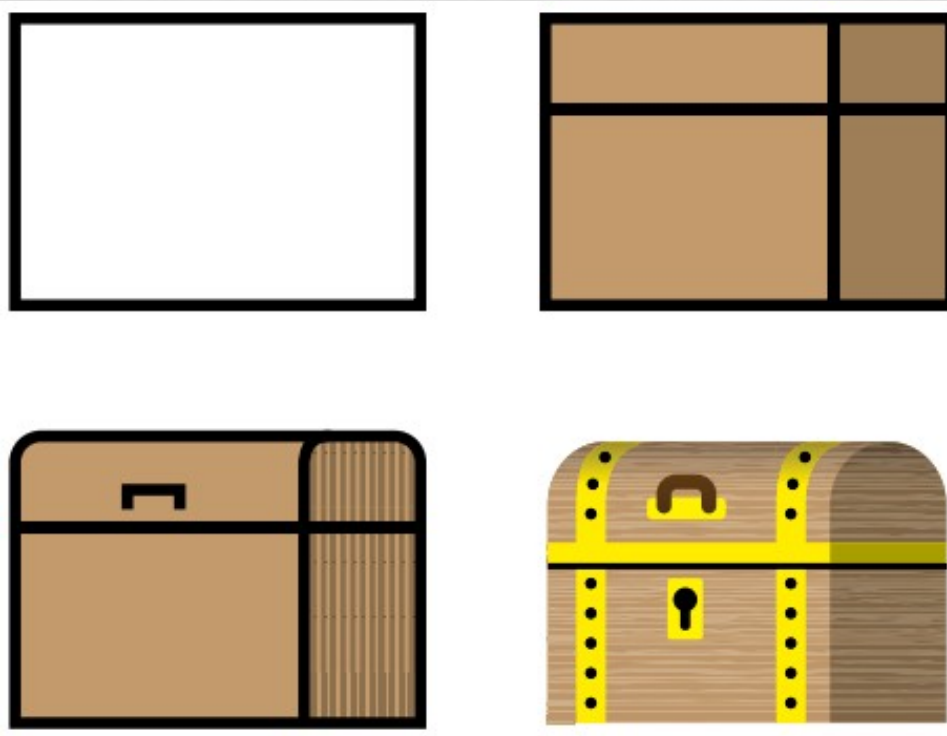
- On which side would you push/pull?



- Environment & Puzzles with appropriate affordance cues draws the player in and makes their interaction & engagement effortless

AFFORDANCE CUES

- Look at how it becomes more and more clear how you should interact with the item as more affordance is added.



AFFORDANCE CUES

- How to implement
 - Beveled Buttons show depth and promote interaction (players want to click them!)
 - Color coding and lightening/darkening sections of your HUD/GUI tells a player which parts they should interact with at a given time.
 - If you want to guide a player from one point to another in a game, put a pathway that's consistent with the style/theme

THE BUSTER PRINCIPLE

- Be kind to your players.
- If it's obvious a player has tried and failed a task several times – make it a bit easier on them.
- The adjustment should be slight – unnoticeable if possible.
- This is a smaller principle in the larger Dynamic Difficulty Adjustment (DDA) field.

COGNITIVE BIASES

- Different people perceive, understand, and react to stimuli (playing games) differently.
- These biases (filters) can skew “what happened” vs. “what I *felt* happened.”
- A lot of these filters tend to operate in similar ways for all people

COGNITIVE BIASES

- One type of Cognitive Bias is **Confirmation Bias**
- People are drawn to information that aligns with what they already believe.
- In games, people are more likely to notice events that confirm their preconceived notions about characters, events, or occurrences.

COGNITIVE BIASES

- The Kuleshov Effect
- Audiences reported the man's face on the left showed a different expression when placed next to different images.



COGNITIVE BIASES

Kahneman and Tversky's "Outbreak"

There is an outbreak of a virus that affects 600 people. You have to select which one of two experimental treatments to administer to the entire group. Because you only have the time and resources to administer one solution. The estimates of their efficacy are listed below.

Program A: 200 people will be saved	Group 1:	72% chose A
Program B: A 1/3 probability that 600 people will be saved, and a 2/3 probability that no people will be saved		

Program C: 400 people will die	Group 2:	
Program D: A 1/3 probability that no one will die, and a 2/3 probability that 600 will die		78% chose D

Mathematically, $A = C$ and $B = D$. However, because of the framing of the words "will be saved" vs. "will die", people's perceptions are different enough that they will select a different program.

COGNITIVE BIASES

- So what's all this psychology stuff have to do with designing games?
- Remember, as a game designer, it's your job to create the experience for your players.
- In order to do that, you have to understand their biases and how they're going to react to your game.

SYNERGY

- The whole is greater than the sum of its parts ($1+1=3$)
- In games, this is accomplished by combining parts of gameplay that already exist into a larger, more cohesive (powerful) experience
- Think of synergy like a list of ingredients (basic mechanics) and allowing the player to mix to come up with something better than the individual components.
- This is how you can have two different games with the same mechanics and drastically different experiences.

SYNERGY

- How to implement:
 - Allow players to combine character movements/actions
 - Fighting games – basic melee attacks can combine into more powerful moves
 - Platform games – combining running & jumping or allow those mechanics to interact with the environment in different ways.
 - Crafting
 - Allow players to take basic items from the world and combine them
 - Is Minecraft an example of synergy?